

Chapter 2: Vector Space

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2.1 Vectors in n-Space

Vectors are fundamental objects in Linear Algebra. In this section, we introduce vectors in the n -dimensional space, and some standard vector operations.

2.1.1 n -Vector

An n -vector is an ordered n -tuple of real numbers $u_1, u_2, \dots, u_n$ denoted by

$$(u_1, u_2, \dots, u_n)$$

For example, a 2-vector is given by an ordered pair (u_1, u_2) while a 3-vector is given by an ordered triplet (u_1, u_2, u_3) .

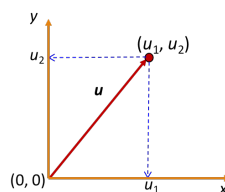
u_i is called the i^{th} component (or i^{th} coordinate) of the n -vector.

When the underlying dimension is clear, we may just refer an n -vector simply as a vector. We should always view an n -vector as a single object. In other words, the n numbers in the vector should be treated collectively. We will use “boldface” letters $\mathbf{u}, \mathbf{v}, \mathbf{w}, \dots$ to represent vectors:

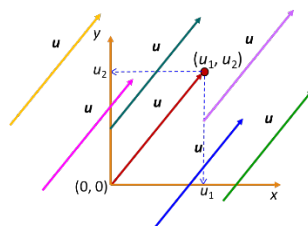
$$\mathbf{u} = (u_1, u_2, \dots, u_n), \quad \mathbf{v} = (v_1, v_2, \dots, v_n).$$

2.1.2 2-vector and 3-vector

Geometrically, a 2-vector $\mathbf{u} = (u_1, u_2)$ is represented by an arrow in the xy -plane.



Note that the red arrow above is not the only arrow that represent $\mathbf{u}$. Any other arrow with the same direction and length as the red one also represent $\mathbf{u}$. There are infinitely many of them.



We can construct an arrow to represent 3-vectors in the xyz-space in a similar way. For general n -vector, we can't explicitly draw it as an arrow, but sometime it might still help to think of them as arrows in order to understand certain concepts regarding vectors.

2.1.3 n -Vectors as Matrices

We can regard an n -vector $(u_1, u_2, \dots, u_n)$ as a matrix with only one row or one column.

More precisely, we call a $1 \times n$ matrix $(u_1 \ u_2 \ \dots \ u_n)$ a row vector

and call an $n \times 1$ matrix $\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$ a column vector.

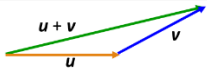
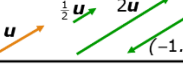
We are free to write vectors in row or column forms, but under certain circumstances, specific form has to be used.

The special n -vector $(0, 0, \dots, 0)$ is called the zero vector, and it correspond to either the $1 \times n$ or $n \times 1$ zero matrix $\mathbf{0}$.

2.1.4 Vector Operations

Here are some basic operations we can perform on n -vectors. They are similar to the matrix operations.

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be n -vectors, and c a real number.

Notation	Algebraic	Geometric
Vector addition $\mathbf{u} + \mathbf{v}$	$(u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$	
Vector subtraction $\mathbf{u} - \mathbf{v}$	$(u_1 - v_1, u_2 - v_2, \dots, u_n - v_n)$	
Negative $-\mathbf{u}$	$(-u_1, -u_2, \dots, -u_n)$	
Scalar multiplication $c\mathbf{u}$	$(cu_1, cu_2, \dots, cu_n)$	

In the case of 2- and 3-vectors, we can also give geometrical interpretation to these operations in terms of the arrows as shown in the last column of the table above.

Vector addition of two vectors: we can move the arrows of the two vectors (without changing directions) so that the end point of one arrow (orange) join with the starting point of the other (blue). Then the sum of the two vectors (green) is obtained by joining the third side of the triangle as shown.

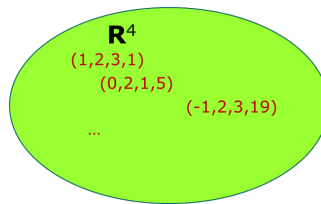
Vector subtraction of two vectors: we move the arrows so that the starting points of the two vectors (orange and blue) coincide. Then we join the third side of the triangle to get the difference (green).

Negative of a vector: It is just represented by the arrow pointing in exactly the opposite direction as the original vector. Both arrows have the same length.

Scalar multiplication of a vector by a scalar (real number): This has the effect of changing length of the arrow while preserving the direction. For example, scalar multiplication by 2 will double the length of the arrow, while a scalar $\frac{1}{2}$ will shorten the length by half; and a negative scalar will reverse the direction of the arrow at the same time.

2.1.5 n-Space

For a fix number n , the set or collection of all the n -vectors is called the n -space, and is represented by the notation $\mathbf{R}^n$. For example, if we use the Venn diagram to represent the 4-space, then its elements are all the 4-vectors such as $(1, 2, 3, 1)$, $(0, 2, 1, 5)$ and so on.



For the 2-space $\mathbf{R}^2$, we can visualize it as the set of all 2-vectors in the xy -plane. For the 3-space $\mathbf{R}^3$, we can view it as the set of all 3-vectors in the xyz -space.

n -spaces are examples of vector spaces. We will see more examples in section 2.3.

The following are different ways to say the same thing:

- $\mathbf{u} \in \mathbf{R}^n$
- $\mathbf{u}$ is an n -vector
- $\mathbf{u} = (u_1, u_2, \dots, u_n)$ for some real numbers $u_1, u_2, \dots, u_n$.

2.1.6 Solution Set as a Subset of $\mathbf{R}^n$

We shall see how the solutions of a linear system is related to the n -space.

$$\begin{cases} x + y + z = 0 \\ x - y + 2z = 1 \end{cases}$$

For this linear system, check that a particular solution is given by

$$x = 2, y = -1 \text{ and } z = -1.$$

Writing in matrix form, this solution can be given by the 3×1 column vector $\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$.

Therefore, we can regard a solution of this system as a 3-vector in $\mathbf{R}^3$.

In fact, this system has infinitely many solutions with general solutions

$$\begin{cases} x = 0.5 - 1.5t \\ y = -0.5 + 0.5t \\ z = t \end{cases}$$

where t is a parameter.

Once again, this can be put in matrix form

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{1}{2} - \frac{3}{2}t \\ -\frac{1}{2} + \frac{1}{2}t \\ t \end{pmatrix}$$

In other words, the set of all the solutions of the above system is a subset of $\mathbf{R}^3$ and contains all 3-vectors given by expression on the right.

In general, if a linear system has n variables, then its solutions are n -vectors, and the solution set is a subset of $\mathbf{R}^n$.

2.1.7 Dot Product

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$, $\mathbf{v} = (v_1, v_2, \dots, v_n)$ vectors in $\mathbf{R}^n$. The dot product of $\mathbf{u}$ and $\mathbf{v}$ is defined to be the number (scalar):

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

The left hand side is the notation of the dot product operation. The right hand side is the sum of the product of the corresponding components of the two vectors, and note that the outcome is a number, not a vector.

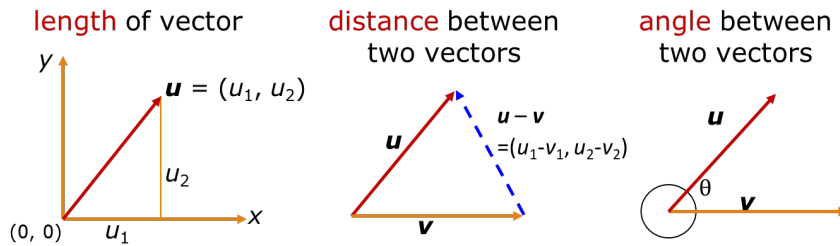
We can perform dot product of a vector with itself, and this will result in the sum of squares of all the components of the vector:

$$\mathbf{u} \cdot \mathbf{u} = u_1^2 + u_2^2 + \dots + u_n^2.$$

2.1.8 Length, Distance and Angles in $\mathbf{R}^2$

We will look at a few geometrical concepts associated with n-vectors, and illustrate how we can apply the dot product in the process, beginning with 2-vectors in $\mathbf{R}^2$.

The notion of length, distance and angles are defined in the intuitive way.



The length of a vector $\mathbf{u} = (u_1, u_2)$ simply means the length of the arrow representing the vector. Using Pythagoras theorem, we easily obtain the length

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2}.$$

The distance between the two vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ refers to the distance between the end points of the arrows, which is given by the length of the vector $\mathbf{u} - \mathbf{v}$:

$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}.$$

The angle between $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ refers to the angle θ between the two arrows representing the vectors. If we use the cosine rule ($a^2 = b^2 + c^2 - 2bc \cos \theta$), we can derive

$$\cos \theta = \frac{u_1 v_1 + u_2 v_2}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

and hence the angle θ (between 0 and π) can be obtained:

$$\theta = \cos^{-1} \left(\frac{u_1 v_1 + u_2 v_2}{\|\mathbf{u}\| \|\mathbf{v}\|} \right).$$

The same notation applies to 3-vectors in $\mathbf{R}^3$. In this case, the algebraic expression for length, distance and angles will involve the 3 components of the vectors instead of two components for 2-vectors.

2.1.9 Length, Distance and Angles in $\mathbf{R}^n$

We can extend the algebraic expression for length, distance and angles to any n -vectors in $\mathbf{R}^n$ in the obvious way.

The length of an n -vector $\mathbf{u} = (u_1, u_2, \dots, u_n)$:

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}.$$

The distance between $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$:

$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}.$$

The angle θ between $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$:

$$\theta = \cos^{-1} \left(\frac{u_1 v_1 + u_2 v_2 + \dots + u_n v_n}{\|\mathbf{u}\| \|\mathbf{v}\|} \right).$$

We can represent these expressions more concisely in terms of dot product.

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}, \quad \|\mathbf{u} - \mathbf{v}\| = \sqrt{(\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})}, \quad \theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \right).$$

For a general n -vector, the length is also known as the norm.

So a vector of norm 1, is a vector with length 1. Such a vector is called a unit vector:

$$\mathbf{u} \text{ is a unit vector} \Leftrightarrow \mathbf{u} \text{ has norm } 1 \Leftrightarrow \|\mathbf{u}\| = 1$$

Example

Let $\mathbf{u} = (1, 2, 0, 1)$, $\mathbf{v} = (1, -1, 1, 1)$ be vectors in $\mathbf{R}^4$.

The norm (length) of $\mathbf{u}$ is given by: $\|\mathbf{u}\| = \sqrt{1^2 + 2^2 + 0^2 + 1^2} = \sqrt{6}$.

The distance between $\mathbf{u}$ and $\mathbf{v}$ is given by: $\|\mathbf{u} - \mathbf{v}\| = \sqrt{(1-1)^2 + (2+1)^2 + (0-1)^2 + (1-1)^2} = \sqrt{10}$.

The angle between $\mathbf{u}$ and $\mathbf{v}$ is given by: $\theta = \cos^{-1} \left(\frac{1 \times 1 + 2 \times (-1) + 0 \times 1 + 1 \times 1}{\sqrt{6} \times \sqrt{4}} \right) = \cos^{-1}(0) = \frac{\pi}{2}$.

We observe that the quotient turns out to be zero, and hence the angle in this case is 90° .

Although we cannot visualize general n -vectors, they still represent certain quantitative objects. The abstract notions of length, distance and angles allow us to make comparison between them. For example, whether one n -vector gives a good approximation for another n -vector. We shall see this in a later section.

2.1.10 Orthogonal Vectors

Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in $\mathbf{R}^n$. If $\mathbf{u} \cdot \mathbf{v} = 0$, we say $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

In this case, the angle between $\mathbf{u}$ and $\mathbf{v}$:

$$\theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \right) = \cos^{-1}(0) = \frac{\pi}{2}.$$

So $\mathbf{u}$ and $\mathbf{v}$ are perpendicular. For example, we can easily check that $\mathbf{u} = (1, 2, 0, 1)$ and $\mathbf{v} = (1, -1, 1, 1)$ has dot product $\mathbf{u} \cdot \mathbf{v} = 0$, and hence are orthogonal to each other in $\mathbf{R}^4$.

2.2 Linear Combination and Linear Span

The topics of linear combinations and linear spans are fundamental in the theory of vector spaces. Many important concepts about n-vectors require the notion of linear combination, which is a combination of scalar multiplication and vector addition applied on a given set of vectors. For a fixed set of n-vectors, we can use linear combinations to generate many more n-vectors. This will lead to the concept of linear span.

2.2.1 Linear Combinations

To talk about linear combination, we must always be given a set of n-vectors to begin with:

$$\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k : \text{a fixed set of vectors in } \mathbf{R}^n$$

For each of these vectors, we are going to multiply by a corresponding scalar: $c_1, c_2, \dots, c_k$:

$$c_1\mathbf{u}_1, c_2\mathbf{u}_2, \dots, c_k\mathbf{u}_k: \text{scalar multiples of the vectors.}$$

Then we add all these scalar multiples of the vectors together to get:

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k.$$

This is called a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$. It is also an n-vector in $\mathbf{R}^n$.

For example, take $\mathbf{u}_1 = (2, 1, 0)$ $\mathbf{u}_2 = (-3, 0, 1)$ and $c_1 = 1, c_2 = 1$. Then we get a specific linear combination:

$$1(2, 1, 0) + 1(-3, 0, 1) = (-1, 1, 1).$$

For the same vectors, if we use general scalars $c_1 = s, c_2 = t$, then we will have a general linear combination with parameters s and t :

$$s(2, 1, 0) + t(-3, 0, 1).$$

Example

Let $\mathbf{u}_1 = (2, 1, 3)$, $\mathbf{u}_2 = (1, -1, 2)$ and $\mathbf{u}_3 = (3, 0, 5)$ be three fixed 3-vectors in $\mathbf{R}^3$.

(a) $\mathbf{v} = (3, 3, 4)$ is a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$
as $(3, 3, 4)$ can be expressed as $a(2, 1, 3) + b(1, -1, 2) + c(3, 0, 5)$. (Why?)

(b) $\mathbf{w} = (1, 2, 4)$ is not a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$
as $(1, 2, 4)$ cannot be expressed as $a(2, 1, 3) + b(1, -1, 2) + c(3, 0, 5)$. (Why?)

We shall see below how to get these answers.

For part (a), we start by writing the given vector $\mathbf{v}$ as a general linear combination $\mathbf{v} = a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_3$

$$(3, 3, 4) = a(2, 1, 3) + b(1, -1, 2) + c(3, 0, 5) \quad (*)$$

where a, b, c are unknown scalars to be determined. Equating the components on the left and right hand side, we get a linear system:

$$\begin{cases} 2a + b + 3c = 3 \\ a - b + 0c = 3 \\ 3a + 2b + 5c = 4 \end{cases}$$

The first equation come from equating the first component on both sides of (*). Similarly, the second and third equations come from the second and third components of the vectors in (*).

So the original equation (*) is equivalent to a linear system, and we can solve this system for the values of a, b, c.

Before we solve the system, we remark that by expressing the vectors in equation (*) in column form, we get:

$$\begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} = a \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + b \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + c \begin{pmatrix} 3 \\ 0 \\ 5 \end{pmatrix}$$

This is called the vector equation form of the above linear system.

We now proceed to solve the linear system by Gaussian elimination, starting from the augmented matrix, we reduce it to this row echelon form:

$$\left(\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 1 & -1 & 0 & 3 \\ 3 & 2 & 5 & 4 \end{array} \right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 0 & -\frac{3}{2} & -\frac{3}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Since the last column is not a pivot column, the system is consistent. This means solution for a, b, c exists. At this point, without finding the solution yet, we can conclude the vector (3, 3, 4) is a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.

If we want to explicitly write this vector as a specific linear combination, we need to go one step further to solve the system. From the row echelon form above, we have two equations:

$$\begin{aligned} 2a + b + 3c &= 3 \\ -\frac{3}{2}b - \frac{3}{2}c &= \frac{3}{2} \end{aligned}$$

By setting the variable c as free parameter t and using back substitution, we get the general solution for the system:

$$\begin{aligned} a &= 2 - t \\ b &= -1 - t \\ c &= t \end{aligned}$$

This means there are many possible values for a, b and c. For example, let $t = 0$, we get $a = 2, b = -1, c = 0$ and hence we have explicitly

$$(3, 3, 4) = 2\mathbf{u}_1 - \mathbf{u}_2$$

(the scalar attached to $\mathbf{u}_3$ is 0, so we do not need to write it out).

If we take $t = 1$ instead, we will get another set of values $a = 1, b = -2, c = 1$ and hence a different linear combination for (3,3,4):

$$(3, 3, 4) = \mathbf{u}_1 - 2\mathbf{u}_2 + \mathbf{u}_3.$$

For part (b): $\mathbf{w} = (1, 2, 4)$.

Like in part (a), we set up the linear combination $\mathbf{w} = a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_3$:

$$(1, 2, 4) = a(2, 1, 3) + b(1, -1, 2) + c(3, 0, 5)$$

The vector equation is equivalent to a linear system

$$\begin{cases} 2a + b + 3c = 1 \\ a - b + 3c = 2 \\ 3a + 2b + 5c = 4 \end{cases}$$

which can be established by comparing components on both sides. We again apply Gaussian elimination to the augmented matrix and get the row echelon form.

$$\left(\begin{array}{ccc|c} 2 & 1 & 3 & 1 \\ 1 & -1 & 0 & 2 \\ 3 & 2 & 5 & 4 \end{array}\right) \xrightarrow{\text{Gaussian Elimination}} \left(\begin{array}{ccc|c} 2 & 1 & 3 & 1 \\ 0 & -\frac{3}{2} & -\frac{3}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 3 \end{array}\right)$$

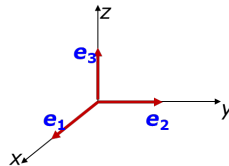
This time we see that the last column is a pivot column, which leads to a row highlighted in pink in the row echelon form. Hence we know the system has no solution and we conclude that the vector (1, 2, 4) is not a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.

2.2.2 Standard Basis Vectors

The standard basis vectors in $\mathbf{R}^3$ is given by the three vectors

$$\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0), \mathbf{e}_3 = (0, 0, 1).$$

Note that these three vectors are the directional unit vectors along the x-, y- and z-axis respectively.



Observe that we can write every vector in $\mathbf{R}^3$ as a linear combination of these standard basis vectors.

Take a general 3-vector $(x \ y \ z)$ in $\mathbf{R}^3$, where x, y, z represent some real numbers.

$$\begin{aligned} (x, y, z) &= (x, 0, 0) + (0, y, 0) + (0, 0, z) \\ &= x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1) \\ &= x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3 \end{aligned}$$

We note that (x, y, z) can be broken up as a sum of three vectors on the right hand side. Then, for these three vectors, we can write them as scalar multiples by pulling out the numbers $x \ y \ z$ from the components. This will then give us a linear combination of the standard basis vectors, with scalars x, y, z respectively.

Take a more specific 3-vector (1, 2, 5), the linear combination in terms of the standard basis vectors is $(1, 2, 5) = 1\mathbf{e}_1 + 2\mathbf{e}_2 + 5\mathbf{e}_3$

In general, for any positive integer n , we have the standard basis vectors for $\mathbf{R}^n$:

$$\mathbf{b}_1 = (1, 0, \dots, 0), \mathbf{b}_2 = (0, 1, \dots, 0), \dots, \mathbf{b}_n = (0, 0, \dots, 1)$$

consisting of n similar vectors with zero component everywhere except for one position as shown.

2.2.3 Linear Span

We now look at the set of all possible linear combinations of a fixed set of vectors. Again let's start with a simple example of two vectors (2,1,0) and (-3,0,1).

How many possible linear combinations $s(2, 1, 0) + t(-3, 0, 1)$ are there? Since there are infinitely many possible scalars s and t , there will be infinitely many linear combinations. In other words, the set of all linear combinations of these two vectors contains infinitely many elements.

We shall call this set the linear span of (2,1,0) and (-3,0,1) and we write it as

$$\text{span}\{ (2,1,0), (-3,0,1) \}.$$

This linear span notation is a concise way to describe this infinite set. It should not be mistaken that this is a finite set with only two vectors $(2,1,0)$ and $(-3,0,1)$. Rather, it contains all possible linear combinations $s(2, 1, 0) + t(-3, 0, 1)$ of the two given vectors.

We now generalize this to a set of k vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ in $\mathbf{R}^n$.

The set of all linear combinations of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$:

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k \quad \text{where } c_1, c_2, \dots, c_k \text{ are in } \mathbf{R}.$$

We call this set the linear span of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ and denote it as $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$.

It contains all linear combinations of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$.

This collection of vectors is an example of a vector space. As all the vectors in this set are n -vectors from $\mathbf{R}^n$, we say $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a subspace of $\mathbf{R}^n$. (We will discuss the concept of subspace in more details in the next section.)

You may always translate the term “linear span” into the phrase “the set of all linear combinations”, since they mean exact the same thing.

Note that when we use the term linear span, we must always refer it to a set of vectors, as the term does not have specific meaning if used stand alone.

Let’s revisit our earlier example: $\mathbf{u}_1 = (2, 1, 3)$, $\mathbf{u}_2 = (1, -1, 2)$ and $\mathbf{u}_3 = (3, 0, 5)$.

(a) $\mathbf{v} = (3, 3, 4)$ is a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. So $\mathbf{v} \in \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

(b) $\mathbf{w} = (1, 2, 4)$ is not a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. So $\mathbf{w} \notin \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

2.2.4 Linear Span of Standard Basis Vectors

Recall that every vector in $\mathbf{R}^3$ is a linear combination of the standard basis vectors

$$\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0), \mathbf{e}_3 = (0, 0, 1).$$

So $\text{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ contains every vector in $\mathbf{R}^3$ and hence

$$\text{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbf{R}^3.$$

Similarly for the standard basis vectors in $\mathbf{R}^n$

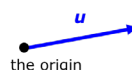
$$\mathbf{b}_1 = (1, 0, \dots, 0), \mathbf{b}_2 = (0, 1, \dots, 0), \dots, \mathbf{b}_n = (0, 0, \dots, 1),$$

we have $\text{span}\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\} = \mathbf{R}^n$.

2.2.5 Linear Span of One Vector

We give some geometrical representation of linear span in the 2 and 3 space. In layman terms the word span means “extend across”.

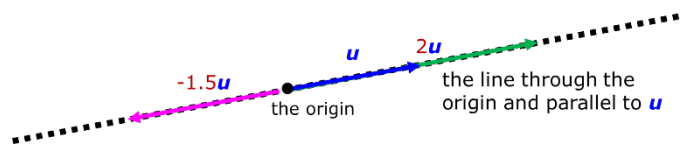
Let $\mathbf{u}$ be a non-zero vector in $\mathbf{R}^2$ or $\mathbf{R}^3$



Since there is only one vector, the linear combinations are just the scalar multiples of $\mathbf{u}$. Hence

$$\text{span}\{\mathbf{u}\} = \text{all scalar multiples } c\mathbf{u}.$$

These are represented by arrows parallel to $\mathbf{u}$ and can be of any length and the arrows can be pointing in the same or opposite direction of $\mathbf{u}$. Hence, this will eventually extend across the entire line parallel to $\mathbf{u}$.

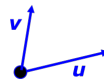


In other words,

$\text{span}\{\mathbf{u}\}$ is represented by the line parallel to $\mathbf{u}$ and contains the origin.

2.2.6 Linear Span of Two Vectors

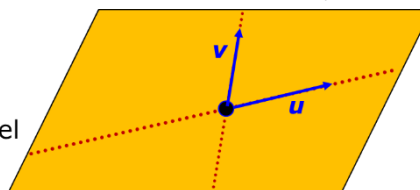
Let $\mathbf{u}, \mathbf{v}$ be two non-parallel vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$



$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \text{all linear combinations } s\mathbf{u} + t\mathbf{v}.$$

For the arrow $\mathbf{u}$, we can extend it across the entire line parallel to $\mathbf{u}$, likewise for $\mathbf{v}$. But do these represent all the linear combinations $s\mathbf{u} + t\mathbf{v}$? The answer is no. These two lines will only contain all the scalar multiples of $\mathbf{u}$ and $\mathbf{v}$ separately, but do not contain the vector addition between $s\mathbf{u}$ and $t\mathbf{v}$. The effect of addition will give arrows that are not on these two lines, but on the plane that contains these two lines as shown.

the plane containing
the origin and parallel
to $\mathbf{u}$ and $\mathbf{v}$



As s and t can be any real numbers, $s\mathbf{u} + t\mathbf{v}$ will give all the arrows that lie on the plane, and hence extend across the entire plane. So

$\text{span}\{\mathbf{u}, \mathbf{v}\}$ is represented by the plane that contains the two vectors $\mathbf{u}$ and $\mathbf{v}$ and the origin.

2.3 Subspaces

In linear algebra, an important technique is to “switch” between matrices and vectors in order to solve problems more effectively.

2.3.1 Switching between vectors and matrices

For example, if we are given a set of 3-vectors $(1,2,0)$, $(3,1,7)$, $(4,4,-2)$, $(0,5,3)$, we may stack them

horizontally to form a 4×3 matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 7 \\ 4 & 4 & -2 \\ 0 & 5 & 3 \end{pmatrix}$. Some time, we may also want to stack them

vertically to form a 3×4 matrix $\mathbf{B} = \begin{pmatrix} 1 & 3 & 4 & 0 \\ 2 & 1 & 4 & 5 \\ 0 & 7 & -2 & 3 \end{pmatrix}$.

Conversely, when we are given a matrix, we may also separate their rows or columns to obtain a set of vectors.

2.3.2 Row Space of a Matrix

Consider the 4×3 matrix

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & -1 & 3 \\ -5 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

The 4 rows of $\mathbf{A}$ can be regarded as 3-vectors in $\mathbf{R}^3$:

$$\begin{array}{l} \text{rows of } \mathbf{A} \\ \mathbf{r}_1 = (2, -1, 0) \\ \mathbf{r}_2 = (1, -1, 3) \\ \mathbf{r}_3 = (-5, 1, 0) \\ \mathbf{r}_4 = (1, 0, 1) \end{array}$$

Now we take the linear span of these vectors:

$$\text{span}\{ (2, -1, 0), (1, -1, 3), (-5, 1, 0), (1, 0, 1) \}.$$

This means we are taking the collection of all linear combinations of the 4 rows $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4$ of matrix $\mathbf{A}$. We call this collection the row space of the matrix $\mathbf{A}$. This is an example of a vector space. Since all the vectors in this row space are from $\mathbf{R}^3$, we say this row space is a subspace of $\mathbf{R}^3$.

- Students tend to have the misconception that the row space consists just the 4 vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4$, which is not true:

$$\text{row space of } \mathbf{A} = \text{span}\{ \mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4 \} \neq \{ \mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4 \}.$$

The row space contains infinitely many vectors, while the right hand side contains only four vectors.

In general, given any matrix, we can construct the row space in a similar way by taking the linear span of all its rows.

2.3.3 Column Space of a Matrix

Column space of a matrix is defined similarly as the linear span of all the columns of the matrix.

Let's use the same matrix $\mathbf{A}$ above.

$$\text{columns of } \mathbf{A} \quad \mathbf{c}_1 = \begin{pmatrix} 2 \\ 1 \\ -5 \\ 1 \end{pmatrix} \quad \mathbf{c}_2 = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} \quad \mathbf{c}_3 = \begin{pmatrix} 0 \\ 3 \\ 0 \\ 1 \end{pmatrix}$$

Now we take the linear span of these column vectors:

$$\text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ -5 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

i.e. we are taking all linear combinations of the 3 columns $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ of matrix $\mathbf{A}$. We call this collection the column space of the matrix $\mathbf{A}$. Again, this is an example of a vector space. This time, the column space is a subspace of $\mathbf{R}^4$. Note that, we usually write the vectors in a column space in column (vertical) form.

Again, do not mistaken the column space as the set with only the 3 column vectors $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$:

$$\text{column space of } \mathbf{A} = \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\} \neq \{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$$

2.3.4 Nullspace of a Matrix

The third type of vector space associated with a given matrix $\mathbf{A}$ is a collection of vectors $\mathbf{u}$ such that when we pre-multiply it with $\mathbf{A}$, the product $\mathbf{A}\mathbf{u} = \mathbf{0}$. This is called the nullspace of a matrix $\mathbf{A}$.

Let us use a different matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 3 & -6 & 9 \end{pmatrix}$$

To find the vectors in the nullspace of $\mathbf{A}$, we set up the matrix equation (on the left below):

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 3 & -6 & 9 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x - 2y + 3z = 0 \\ 2x - 4y + 6z = 0 \\ 3x - 6y + 9z = 0 \end{cases}$$

Note that this is the same as a homogeneous system. Putting the system in standard equation form (on the right side above), we proceed to solve this using Gaussian elimination as usual.

The general solution (we omit the Gaussian elimination here) of the system is given by

$$\text{general solution} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2s - 3t \\ s \\ t \end{pmatrix}$$

In other words, all the vectors in the nullspace of $\mathbf{A}$ are given by the above general solution.

Note that we can rewrite the expression in the general solution by separating it into two vectors in terms of the two parameters s and t (the middle term below). We can further pull out the parameters s and t respectively to get two the rightmost term below.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2s - 3t \\ s \\ t \end{pmatrix} = \begin{pmatrix} 2s \\ s \\ 0 \end{pmatrix} + \begin{pmatrix} -3t \\ 0 \\ t \end{pmatrix} = s \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$$

In other words, we can express all vectors in the nullspace of A as a general linear combination of the two fixed vectors $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$.

So we can write

$$\text{the nullspace of } A = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

So the nullspace is a vector space, and this nullspace is a subspace of $\mathbf{R}^3$.

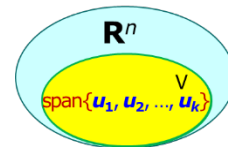
In general, if A is an $m \times n$ matrix, the nullspace of A is a subspace of $\mathbf{R}^n$. Note that we normally write the vectors in the nullspace in column form, so that they are compatible with the matrix multiplication of A .

2.3.5 Subspaces

From the above discussion, when we are given an $m \times n$ matrix A :

- The row space of A is a subspace of $\mathbf{R}^n$;
- The column space of A is a subspace of $\mathbf{R}^m$;
- The nullspace of A is a subspace of $\mathbf{R}^n$.

In general, a collection V of n -vectors is said to be a subspace of $\mathbf{R}^n$ if V can be expressed as a linear span, i.e. there is a given set of n -vectors $u_1, u_2, \dots, u_k$ such that $V = \text{span}\{u_1, u_2, \dots, u_k\}$.



In this course, when we say a set of vectors is a “vector space”, we are always referring to the n -space $\mathbf{R}^n$ (for some n) or any subspace of $\mathbf{R}^n$. In other words, a vector space is a general term that can be used to describe any of the “spaces” that we have discussed so far.

2.3.6 Closure properties

A subspace V satisfies the closure properties:

- (i) for all $u, v \in V$, we must have $u + v \in V$.
- (ii) for all $u \in V$ and $c \in \mathbf{R}$, we must have $cu \in V$.

Note that every subspace of $\mathbf{R}^n$ is a subset of $\mathbf{R}^n$ but not every subset of $\mathbf{R}^n$ is a subspace of $\mathbf{R}^n$. (See the next two examples.)

Example:

Consider the subspace $V = \text{span}\{(1, 0)\}$ of $\mathbf{R}^2$.

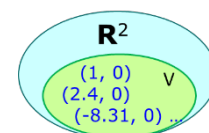
Note that V containing all vectors of the form $(a, 0)$, i.e. V contains 2-vectors $(1, 0)$, $(2.4, 0)$, $(-8.31, 0)$, etc.

Take any two random vectors $(a, 0)$ and $(b, 0)$ in V , we have

$$(a, 0) + (b, 0) = (a+b, 0)$$

which is again a vector in V . So it satisfies closure property under vector addition.

Take a random vector $(a, 0)$ in V and a random scalar c , we have



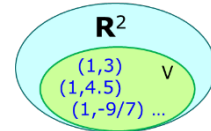
$$c(a, 0) = (ca, 0)$$

which is again a vector in V . So it satisfies closure property under scalar multiplication.

This subspace V is represented by a straight line that passes through the origin in the xy -plane and parallel to the vector $(1,0)$. In fact, any straight line in XY -plane that passes through the origin represent a subspace of the 2-space.

Example:

This next example is a subset of the 2-space which is not a subspace. We will see that the closure properties are not satisfied.



Let V be the subset of $\mathbf{R}^2$ containing all vectors of the form $(1, a)$. i.e. V contains 2-vectors $(1, 3)$, $(1, 4.5)$, $(1, -9/7)$ etc.

Let's take two specific vectors $(1, 3)$ and $(1, 5)$ in V . Then

$$(1, 3) + (1, 5) = (2, 8) \notin V.$$

So the closure property under vector addition is not satisfied.

Let's take a specific vector $(1, 3)$ in V and a specific scalar 4. Then

$$4(1, 3) = (4, 12) \notin V.$$

So the closure property under scalar multiplication is not satisfied.

2.3.7 Solution Space of Homogeneous System

Earlier, we have seen that if $\mathbf{A}$ is an $m \times n$ matrix, the nullspace of $\mathbf{A}$ is a subspace of $\mathbf{R}^n$. In fact this subspace come from the solutions of the homogeneous system $\mathbf{Ax} = \mathbf{0}$ with m equations and n variables. So we have the following:

Let $\mathbf{Ax} = \mathbf{0}$ be a homogeneous system with n variables. Then its solution set is a subspace of $\mathbf{R}^n$. We will also call this solution set a solution space of the homogeneous system.

By the closure properties of space, this means:

If $\mathbf{u}$ and $\mathbf{v}$ are two random solutions of $\mathbf{Ax} = \mathbf{0}$, then we expect $\mathbf{u} + \mathbf{v}$ to be also a solution of $\mathbf{Ax} = \mathbf{0}$.

If $\mathbf{u}$ is a random solution of $\mathbf{Ax} = \mathbf{0}$ and c is a scalar, then we expect $c\mathbf{u}$ to be also a solution of $\mathbf{Ax} = \mathbf{0}$.

In fact, any linear combination $c\mathbf{u} + d\mathbf{v}$ will also be a solution of $\mathbf{Ax} = \mathbf{0}$.

Note that only the solution set of a homogeneous system gives a subspace of $\mathbf{R}^n$. For a non-homogeneous system $\mathbf{Ax} = \mathbf{b}$, the solution set is not a subspace of n -space. In this case, we cannot call the solution set of non-homogeneous system a solution space.

2.4 Linear Independence

We introduce another important concept in vector space. Given a set of n -vectors, sometime it is essential to know whether there is any linear relationship among the vectors. When such a relationship exists, we say the set of vectors are linearly dependent, otherwise, we say the set is linearly independent.

2.4.1 Redundant Vector

Let's look at two separate linear spans: $\text{span}\{(1, 1, 1)\}$ and $\text{span}\{(1, 1, 1), (-1, -1, -1)\}$. Note that the second linear span has one additional "spanning" vector $(-1, -1, -1)$ than the first linear span.

Recall that:

$\text{span}\{(1, 1, 1)\}$ contains all scalar multiples $c(1, 1, 1)$ of $(1, 1, 1)$:

$\text{span}\{(1, 1, 1), (-1, -1, -1)\}$ contains all linear combinations $a(1, 1, 1) + b(-1, -1, -1)$ of the two vectors $(1, 1, 1)$ and $(-1, -1, -1)$.

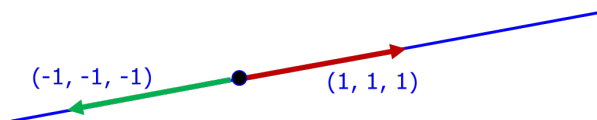
It is easy to see that, regardless of what values we substitute for a and b , the linear combination $a(1, 1, 1) + b(-1, -1, -1)$ can be reduced to a scalar multiple $c(1, 1, 1)$.

Hence, the two linear spans are actually equal:

$$\text{span}\{(1, 1, 1)\} \xleftrightarrow{\text{equal}} \text{span}\{(1, 1, 1), (-1, -1, -1)\}$$

In other words, having the additional vector $(-1, -1, -1)$ in the span of $(1, 1, 1)$ does not change the set of vectors. So we say there is a redundant vector in the span of $(1, 1, 1)$ and $(-1, -1, -1)$.

We can visualize this geometrically. The vector $(1, 1, 1)$ is represented by an arrow.



The linear span of this vector extend across to the entire line that are parallel to this arrow.

Now $(-1, -1, -1)$ is the negative of $(1, 1, 1)$ and hence is parallel to $(1, 1, 1)$ in the opposite direction.

So the linear span of the two vectors will just extend across to the same line, and hence representing the same set.

Let's look at another example: $\text{span}\{(1, 1, 1), (1, 0, -2)\}$ and $\text{span}\{(1, 1, 1), (1, 0, -2), (2, 3, 5)\}$. Again the second linear span has one additional "spanning" vector $(2, 3, 5)$ than the first linear span.

$\text{span}\{(1, 1, 1), (1, 0, -2)\}$ contains all linear combinations $p(1, 1, 1) + q(1, 0, -2)$;

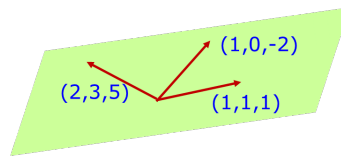
$\text{span}\{(1, 1, 1), (1, 0, -2), (2, 3, 5)\}$ contains all linear combination $a(1, 1, 1) + b(1, 0, -2) + c(2, 3, 5)$.

The two linear spans are the same because $(2, 3, 5)$ is a linear combination of $(1, 1, 1)$ and $(1, 0, -2)$:

$$(2, 3, 5) = 3(1, 1, 1) + (-1)(1, 0, -2)$$

So a linear combination $a(1, 1, 1) + b(1, 0, -2) + c(2, 3, 5)$ can be reduced to a linear combination $p(1, 1, 1) + q(1, 0, -2)$. In other words, having the additional vector $(2, 3, 5)$ in the linear span of $(1, 1, 1)$ and $(1, 0, -2)$ does not change the set of vectors. So $(2, 3, 5)$ can be regarded as a redundant vector in $\text{span}\{(1, 1, 1), (1, 0, -2), (2, 3, 5)\}$.

Let's look at the geometrical interpretation. The two vectors $(1, 1, 1)$ and $(1, 0, -2)$ are represented by two non parallel arrows. The linear span of these two vectors extend across to the entire plane that contains the two arrows.



As we have seen $(2, 3, 5)$ is a linear combination of these two vectors, it must lie on the same plane as the other two vectors. So the linear span of the three vectors will also extend across to the same plane.

2.4.2 Linear Independence and Dependence

Let $u_1, u_2, \dots, u_k$ be a set of vectors in $\mathbb{R}^n$.

Set up the vector equation

$$c_1 u_1 + c_2 u_2 + \dots + c_k u_k = \mathbf{0}.$$

This is equivalent to a homogeneous linear system with variables $c_1, c_2, \dots, c_k$. (See examples below on the vector equation form of a linear system.)

Recall that a homogeneous system has either only the trivial solution (case I) or infinitely many solutions with non-trivial solutions (case II).

Case I: The system has only the trivial solution,

i.e. the only possible scalars are: $c_1 = 0, c_2 = 0, \dots, c_k = 0$

In this case, we say the set of vectors $u_1, u_2, \dots, u_k$ are linearly independent.

Case II: The system has non-trivial solutions,

i.e. there are scalars $c_1, c_2, \dots, c_k$ that are not all zero.

In this case, we say the set $u_1, u_2, \dots, u_k$ are linearly dependent.

Example: Linear Independence

Determine whether the set of vectors $(1, 0, 0, 1)$, $(0, 2, 1, 0)$, $(1, -1, 1, 1)$ are linearly independent.

Set up the vector equation (here we write the vectors in column form):

$$c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 2 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

This is a linear system (in vector equation form) whose augmented matrix can be written down directly (on the left below):

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right] \xrightarrow{\text{G.E.}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Performing Gaussian elimination gives us the row echelon form on the right above. Since there is no non-pivot column in this row echelon form, we know there is exactly one solution, which must be the trivial solution: $c_1 = 0, c_2 = 0, c_3 = 0$. So $(1, 0, 0, 1), (0, 2, 1, 0), (1, -1, 1, 1)$ are linearly independent.

Example: Linear dependence

Determine whether the set of vectors $(1, -2, 3), (5, 6, -1), (3, 2, 1)$ are linearly independent.

Set up the vector equation:

$$c_1 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 5 \\ 6 \\ -1 \end{pmatrix} + c_3 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Again we write down the augmented matrix of this linear system directly (on the left below):

$$\left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ -2 & 6 & 2 & 0 \\ 3 & -1 & 1 & 0 \end{array} \right) \xrightarrow{\text{G.E.}} \left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ 0 & 16 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Performing Gaussian elimination gives us the row echelon form on the right above. Since there is one non-pivot column (third column) in this row echelon form, there are infinitely many solutions for the system, i.e. there exist non-trivial solutions for c_1, c_2, c_3 . So $(1, -2, 3), (5, 6, -1), (3, 2, 1)$ are linearly dependent.

2.4.3 Linear independence and Determinant

There is an alternative way to determine linear independence of a set of vectors when the linear system involved has the same number of equations and variables.

For our previous example, recall that the system

$$c_1 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 5 \\ 6 \\ -1 \end{pmatrix} + c_3 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

has non-trivial solution, and hence the three vectors are linearly dependent. Since there are three vectors, and these vectors come from the 3-space $\mathbf{R}^3$, when we rewrite the system in matrix equation form,

$$\begin{pmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

the coefficient matrix is a 3x3 square matrix, so that we can compute the determinant:

$$\begin{vmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{vmatrix} = 0$$

Since the determinant turns out to be 0, from chapter 1 section 1.6, a homogeneous system $\mathbf{Ax} = \mathbf{0}$ has non-trivial solutions when the coefficient matrix $\mathbf{A}$ is singular, which in turn means $\det(\mathbf{A}) = 0$. This coincides with our observation.

In general, when there are n vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ in $\mathbf{R}^n$, we can form an $n \times n$ matrix $\mathbf{A}$ using $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ (written in column form) as the n columns of the matrix.

- If $\det(\mathbf{A}) = 0$, then $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ are linearly dependent
- If $\det(\mathbf{A}) \neq 0$, then $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ are linearly independent.

2.4.4 Linear Dependence and Redundancy

Let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ be a set with more than 1 vector.

If $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly dependent, then at least one of the vectors $\mathbf{u}_i$ can be written as a linear combination of the other vectors. The converse is also true.

In this case, we know $\mathbf{u}_i$ is a “redundant” vector in the linear span of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$

Examples:

The set of vectors $(1, 1, 1), (1, 0, -2), (2, 3, 5)$ are linearly dependent as $(2, 3, 5)$ is a “redundant” vector in the span of the three vectors.

The set of vectors $(1, 1, 1), (1, 0, -2)$ are linearly independent as there is no “redundant” vector in the span of the two vectors.

2.4.5 Linear Independence of Two Vectors

Let $\mathbf{u}, \mathbf{v}$ be a set with two vectors in $\mathbf{R}^n$.

Suppose $\mathbf{u}, \mathbf{v}$ are linearly dependent. The vector equation $c\mathbf{u} + d\mathbf{v} = \mathbf{0}$ has non-trivial solution (so at least one of c and d is non-zero).

$$\text{- If } c \neq 0, \text{ then } \mathbf{u} = (-d/c)\mathbf{v}$$

$$\text{- If } d \neq 0, \text{ then } \mathbf{v} = (-c/d)\mathbf{u}$$

In either case, we have $\mathbf{u}$ and $\mathbf{v}$ being scalar multiple of each other. So either one of them is a redundant vector between the two.

This gives us a very simple way to decide whether two vectors are linearly independent by inspecting their components:

- If $\mathbf{u}$ and $\mathbf{v}$ are scalar multiples of each other, they are linearly dependent;
- If $\mathbf{u}$ and $\mathbf{v}$ are not scalar multiples of each other, they are linearly independent.

2.4.6 A Condition for Linear Dependence

Let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ be a set of k vectors in $\mathbf{R}^n$.

If $k > n$, then the set of vectors are linearly dependent.

Let's look at some examples.

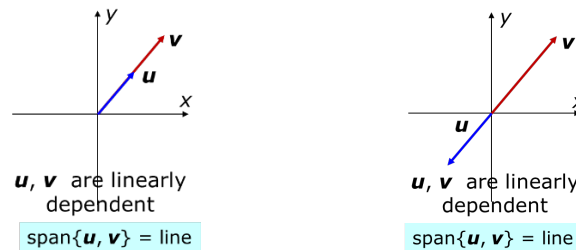
1. In $\mathbf{R}^2$, a set of three or more vectors must be linearly dependent.
So we know immediately that $(1,2), (3,4), (5,6)$ are linearly dependent
2. In $\mathbf{R}^3$, a set of four or more vectors must be linearly dependent.
We have $(1,2,3), (3,4,5), (5,6,7), (7,8,9)$ are linearly dependent

Note that the converse of this result does not hold. In other words, if $k < n$, it does not mean the k vectors are linearly independent. For example, in the 3-space, the two vectors $(1 \ 1 \ 1)$ and $(-1 \ -1 \ -1)$ are linearly dependent, as they are scalar multiple of each other.

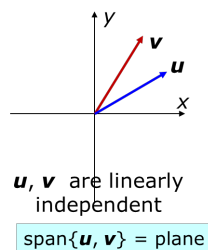
2.4.7 Geometrical Meaning: Two Vectors

In $\mathbf{R}^2$ (or $\mathbf{R}^3$), two vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent if they lie on the same line.

The two arrows representing the vectors either point in the same direction or in exact opposite direction. In both cases, the two arrows lie on the same line as shown below.



Two vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly independent if they do not lie on the same line.



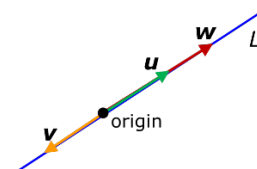
For the last case, the linear span of the two vectors is the plane that contains both vectors. Note that, in $\mathbf{R}^2$, there is only one plane, which is the xy -plane itself, representing the entire $\mathbf{R}^2$. This means:

when $\mathbf{u}$ and $\mathbf{v}$ are linearly independent vectors in $\mathbf{R}^2$, then $\text{span}\{\mathbf{u}, \mathbf{v}\} = \mathbf{R}^2$.

2.4.8 Geometrical Meaning: Three Vectors

In $\mathbf{R}^2$ (or $\mathbf{R}^3$), three vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent if they lie on the same line or same plane.

Case 1: When all three arrows are parallel and starting from the origin, either pointing in the same or opposite direction, then they lie on the same line, and hence the three vectors are linearly dependent.

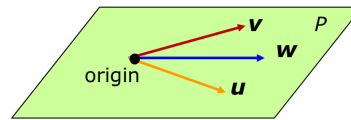


In this case the linear span of the three vectors is represented by the line L that the three vectors lie on. In fact, there are two redundant vectors among $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, since we know the linear span of a single vector will give the same line.

$$\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{span}\{\mathbf{u}\} = \text{span}\{\mathbf{v}\} = \text{span}\{\mathbf{w}\} = L$$

So we can remove any two of the three vectors from the linear span, and they will all be equal to each.

Case 2: When the three arrows are not all parallel to each other but they all lie on the same plane.



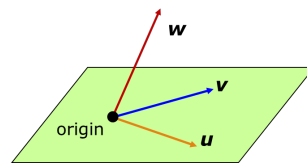
The three vectors are again linearly dependent. In this case the linear span of the three vectors is represented by the plane P that the three vectors lie on. This time, there is only one redundant vectors among $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, since we know the linear span of two non parallel vectors will give the same plane.

$$\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{span}\{\mathbf{u}, \mathbf{v}\} = \text{span}\{\mathbf{u}, \mathbf{w}\} = \text{span}\{\mathbf{v}, \mathbf{w}\} = P.$$

So, if the three vectors are all not parallel to each other, we can remove any one of the three vectors from the linear span, and they will all be equal to each.

Case 3: This case only applies to $\mathbf{R}^3$, when the arrows representing the three vectors does not lie on any common plane. (Note that in $\mathbf{R}^2$, every vector must lie on the xy -plane. So this scenario cannot happen)

In other words, when we try to find a plane that contains two of the three arrows, the remaining one will always be sticking out from the plane, as shown in the diagram.



In this case, the three vectors are linearly independent.

The linear combination of $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ will generate all the 3-vectors in the 3-space. Hence

when $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are linearly independent vectors in $\mathbf{R}^3$, then $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \mathbf{R}^3$.

This means there is no redundant vector among the three. If we try to remove any one of the vectors among $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$, the linear span of the remaining two vectors will be different.

So none of $\text{span}\{\mathbf{u}, \mathbf{v}\}$, $\text{span}\{\mathbf{u}, \mathbf{w}\}$, $\text{span}\{\mathbf{v}, \mathbf{w}\}$ is equal to $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.

2.5 Basis and Dimension

A vector space consists of infinitely many vectors. We can find a set with only a finite number of vectors that can generate all the vectors in a given vector space. Such a set with the smallest possible number of vectors is called a basis for the vector space., and the dimension of a vector space is defined to be the number of vectors in a basis of the space.

2.5.1 Standard Basis Vectors in $\mathbf{R}^3$

We begin with the standard basis vectors in $\mathbf{R}^3$:

$$\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0), \mathbf{e}_3 = (0, 0, 1).$$

We have seen earlier that

$$\diamond \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbf{R}^3$$

as any 3-vector in $\mathbf{R}^3$ can be written as a linear combination of these standard basis vectors. So we can regard these three vectors as the building block that generate all the vectors in $\mathbf{R}^3$.

On the other hand, we also have

$$\diamond \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \text{ are linearly independent}$$

This means there is no redundant vector among $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.

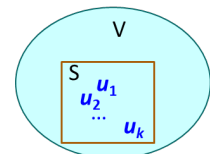
Let us denote this set by $S = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. Since there is no redundant vector in S , if we try to remove any one of the vectors from S , the linear span will no longer be the entire $\mathbf{R}^3$. So we say S is a smallest possible set of vectors that generate every vector in $\mathbf{R}^3$. Such a smallest possible set is called a basis for $\mathbf{R}^3$.

2.5.2 Basis and Dimension

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a subset of a vector space V . (Recall that a vector space can either be $\mathbf{R}^n$ or any subspace of the $\mathbf{R}^n$.) Then

S is called a basis for V if

1. $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\} = V$; and
2. $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent.



In other words, S is a smallest possible set of vectors in V that generate every vector in V .

The number of vectors in a basis for V is called the dimension of V and is denoted by $\dim V$. In other words, if $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a basis for V , then $\dim V = k$. It is the smallest possible number of vectors that can generate V .

Example: Basis for $\mathbf{R}^3$

We have seen that the standard basis vectors $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a basis for $\mathbf{R}^3$.

So $\dim \mathbf{R}^3 = 3$.

A vector space can have many different bases, but they always have the same dimension.

In other words, there are many different ways to form building blocks (with the same number) for the vector space.

For example, in $\mathbf{R}^3$, it is not difficult to see that:

$\{ (2, 0, 0), (0, 2, 0), (0, 0, 2) \}$ is a basis for $\mathbf{R}^3$ (multiplying the standard basis vectors by scalar 2, will still give a building block for $\mathbf{R}^3$).

$\{ (1, 2, 1), (2, 9, 0), (3, 3, 4) \}$ is another basis for $\mathbf{R}^3$. This one is not so obvious. We shall now check the this is a basis by showing the set is linearly independent (see below*):

We set up the vector equation (linear system):

$$c_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 9 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and perform Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 9 & 3 & 0 \\ 1 & 0 & 4 & 0 \end{array} \right) \xrightarrow{\text{G. E.}} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & -\frac{1}{5} & 0 \end{array} \right)$$

Since there is no non-pivot column in the row echelon form, the system has only trivial solution. So $(1, 2, 1), (2, 9, 0), (3, 3, 4)$ are linearly independent and we conclude that $\{ (1, 2, 1), (2, 9, 0), (3, 3, 4) \}$ is a basis for $\mathbf{R}^3$.

* Any set of three linearly independent 3-vectors is a basis for $\mathbf{R}^3$. More generally, any set of n linearly independent n -vectors is a basis for $\mathbf{R}^n$.

2.5.3 Basis for Row Space

Let's start with an example:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix}$$

To find a basis for the row space and column space, we need to look at the row echelon form $\mathbf{R}$.

$$\mathbf{R} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \\ \\ \\ \end{matrix}$$

R.E.F

There are three non-zero rows $\mathbf{v}_1 \mathbf{v}_2 \mathbf{v}_3$. These three nonzero rows are obtained by performing row operations on the original rows of $\mathbf{A}$, and hence they are linear combinations of the rows of $\mathbf{A}$. This means they belong to the row space of $\mathbf{A}$. In fact, $\mathbf{v}_1 \mathbf{v}_2 \mathbf{v}_3$ forms a basis for the row space of $\mathbf{A}$.

A basis for the row space of $\mathbf{A}$ is given by $\{ (1,2,2,1), (0,1,1,1), (0,0,1,1) \}$.

Note that this is not the only basis for the row space of $\mathbf{A}$. For example, the non-zero rows of the reduced row echelon form of $\mathbf{A}$ will give another basis for the row space.

In general,

the non-zero rows in any r.e.f. R of a matrix $A \rightarrow$ a basis for the row space of A .

The dimension of the row space will therefore be given by the number of nonzero rows of the row echelon form. If you recall from the earlier chapter, this is the rank of the matrix:

$$\dim(\text{row space of } A) = \text{rank}(A).$$

2.5.4 Basis for Column Space

Column space

We look at the pivot columns of the row echelon form.

$$R = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

pivot columns

However, these pivot columns are not linear combinations of the columns in the original matrix A , and hence they may not belong to the column space of A . Nevertheless,

the columns of A corresponding to pivot columns of $R \rightarrow$ a basis for the column space of A .

In our example, since the pivot columns of the row echelon form R are the first three columns, we will look at the corresponding first three columns of the original matrix A .

$$A = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix}$$

$c_1 \quad c_2 \quad c_3$

These three columns $c_1 \ c_2 \ c_3$ form a basis for the column space of A .

A basis for the column space of A is given by $\left\{ \begin{pmatrix} 1 \\ 3 \\ 4 \\ -2 \\ 5 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 6 \\ 9 \\ -1 \\ 8 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 6 \\ 9 \\ -1 \\ 9 \\ 7 \end{pmatrix} \right\}$. Again, there are many

other possible bases for the column space that can be obtained using other approach.

The dimension of the column space of A will be given by the number of pivot columns of the row echelon form. This number is again the rank of A .

$$\dim(\text{column space of } A) = \text{rank}(A).$$

In other words, the dimension of the row space and the column space of a matrix are the same,

$$\dim(\text{row space of } A) = \dim(\text{column space of } A)$$

and they are both equal to the rank of the matrix.

2.5.5 Basis for Linear Span

Let $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ where $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are some vectors in $\mathbf{R}^n$. So V is a subspace of $\mathbf{R}^n$.

If $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is linearly independent, then it is a basis for V .

If $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is linearly dependent, then there are redundant vectors among them. We need to remove the redundant vectors to get a basis for V .

To identify and remove redundant vectors, we can use the “column space” method:

- Form a matrix $\mathbf{A}$ using $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ as its columns.
- Reduce $\mathbf{A}$ to a row echelon form $\mathbf{R}$
- Identify the pivot columns in $\mathbf{R}$
- The corresponding columns in $\mathbf{A}$ form a basis for V .

Example:

Let $V = \text{span}\{(1,2,2,1), (3,6,6,3), (4,9,9,5), (-2,-1,-1,1), (5,8,9,4)\}$, which is a subspace of $\mathbf{R}^4$.

We form the matrix $\mathbf{A} = \begin{pmatrix} 1 & 3 & 4 & -2 & 5 \\ 2 & 6 & 9 & -1 & 8 \\ 2 & 6 & 9 & -1 & 9 \\ 1 & 3 & 5 & 1 & 4 \end{pmatrix}$ and perform G.E. to obtain a r.e.f.

$$\mathbf{R} = \begin{pmatrix} 1 & 3 & 4 & -2 & 5 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

As the 2nd and 4th columns of $\mathbf{R}$ are non-pivot columns, the corresponding 2nd and 4th column of $\mathbf{A}$ are the redundant vectors in the linear span. On the other hand, the 1st, 3rd and 5th columns of $\mathbf{R}$ are pivot columns, so the corresponding 1st, 3rd and 5th columns of $\mathbf{A}$ will form a basis for the linear span.

Hence we can conclude that $\{(1,2,2,1), (4,9,9,5), (5,8,9,4)\}$ is a basis for V .

2.5.6 Basis for Solution Space

Find a basis for and determine the dimension of the solution space of the homogeneous system

$$\begin{cases} 2v + 2w - x + z = 0 \\ -v - w + 2x - 3y + z = 0 \\ x + y + z = 0 \\ v + w - 2x - z = 0 \end{cases}$$

This is a system with 5 variables, and hence its solution space is a subspace of $\mathbf{R}^5$. After performing Gaussian elimination, the general solution is given on the right side below:

$$\begin{pmatrix} v \\ w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -s-t \\ s \\ -t \\ 0 \\ t \end{pmatrix} = s \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

Like before, we can rewrite this expression as a general linear combination of some fixed vectors, as given on the right side above. Let's denote the two vectors in the linear combination by:

$$\mathbf{u}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

So the solution space = $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$. This is one of the conditions that $\mathbf{u}_1, \mathbf{u}_2$ is a basis for the solution space. The vectors $\mathbf{u}_1, \mathbf{u}_2$ obtained in such a way are always linearly independent. (In this case, observe that these two vectors are not scalar multiplication of each other.)

So we can conclude that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is a basis for the solution space of the given system, and

$$\dim(\text{solution space}) = 2.$$

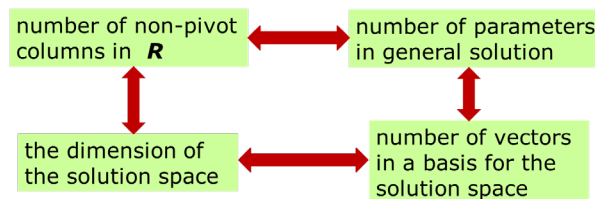
We observe that:

$\dim(\text{solution space}) = \text{number of parameters in the general solution.}$

This is true in general.

Dimension of Solution Space

Given any homogeneous system, to find a basis and dimension for the solution space, the standard approach is to perform Gaussian elimination to get the row echelon form $\mathbf{R}$.



The number of non-pivot columns in the row echelon form corresponds to the number of parameters in the general solution. This number in turn corresponds to the number of vectors in a basis for the solution space. By definition, this is the dimension of the solution space. Hence the four numbers in the diagram above are all the same.

2.5.7 Basis for Nullspace

Recall that the nullspace of a matrix $\mathbf{A}$ is the solution space of the homogeneous system $\mathbf{Ax} = \mathbf{0}$. So

- a basis for the nullspace of $\mathbf{A}$ is same as a basis for the solution space of $\mathbf{Ax} = \mathbf{0}$.
- the dimension for the nullspace of $\mathbf{A}$ is same as the dimension for the solution space of $\mathbf{Ax} = \mathbf{0}$.

The dimension of the nullspace of $\mathbf{A}$ is also called the nullity of $\mathbf{A}$, and it is denoted by $\text{nullity}(\mathbf{A})$.

Example: To find the nullspace and nullity of

$$\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & -2 & 0 & -1 \end{pmatrix},$$

we look at the homogeneous system $\mathbf{Ax} = \mathbf{0}$

$$\begin{cases} 2v + 2w - x + z = 0 \\ -v - w + 2x - 3y + z = 0 \\ x + y + z = 0 \\ v + w - 2x - z = 0 \end{cases}$$

Since we know from earlier discussion that a basis for this system is given by

$$\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

so it is also a basis for the nullspace of $\mathbf{A}$, and $\text{nullity}(\mathbf{A}) = 2$.

2.5.8 Dimension Theorem for Matrices

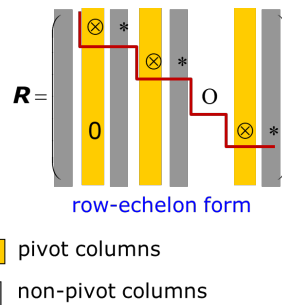
If $\mathbf{A}$ is a matrix with n columns, then $\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n$.

Recall that

$$\text{rank}(\mathbf{A}) = \dim(\text{column space of } \mathbf{A}) \quad \text{and} \quad \text{nullity}(\mathbf{A}) = \dim(\text{nullspace of } \mathbf{A}).$$

That explains why this result is called the Dimension Theorem for Matrices.

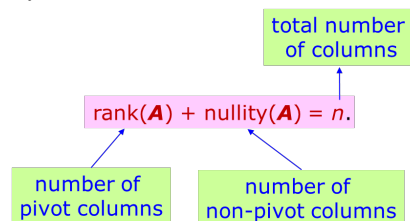
We look at the row echelon form $\mathbf{R}$ of the matrix $\mathbf{A}$.



The pivot columns correspond to a basis for the column space of $\mathbf{A}$, so
the number of pivot columns = $\text{rank}(\mathbf{A})$.

On the other hand, the non-pivot columns correspond to the parameters in general solutions of the homogeneous system $\mathbf{Ax} = \mathbf{0}$, which in turn corresponds to a basis for the nullspace of $\mathbf{A}$. so
the number of non-pivot columns = $\text{nullity}(\mathbf{A})$.

The following diagram explains why the dimension theorem of matrices works.



2.6 Coordinate Vectors

In the previous section, we introduced the basis for a vector space, which form the building blocks for the space. In this section, we shall see how a basis provides a coordinate system and serves as a unit of measurement for the vectors in the vector space.

2.6.1 Uniqueness Expression in terms of Basis

Let $S = \{u_1, u_2, \dots, u_k\}$ be a basis for a vector space V . Then

every vector v in V can be expressed in the form $v = c_1u_1 + c_2u_2 + \dots + c_ku_k$ in exactly one way.
i.e. there is only one possible set of values for $c_1, c_2, \dots, c_k$.

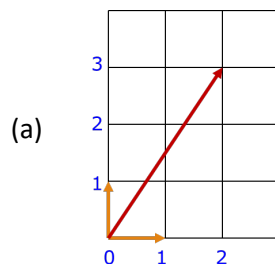
The first part of the property “can be expressed in the form $v = c_1u_1 + c_2u_2 + \dots + c_ku_k$ ” follows from the fact that the basis spans the entire V .

The second part “exactly one way” is a consequence that the basis is linearly independent. What this means is that, different linear combinations of the basis vectors will give different vectors in the space V .

For example, suppose $\{u_1, u_2, u_3\}$ is a basis for $\mathbb{R}^3$. If the linear combination $3u_1 + 5u_2 + 2u_3 = v$ and the linear combination $2u_1 + 4u_2 + 6u_3 = w$, then $v \neq w$, as the two linear combinations have different coefficients. (Recall that, it is possible that two different linear combinations of a set of vectors can be equal to the same vector. This will happen when the set of vectors are linearly dependent.)

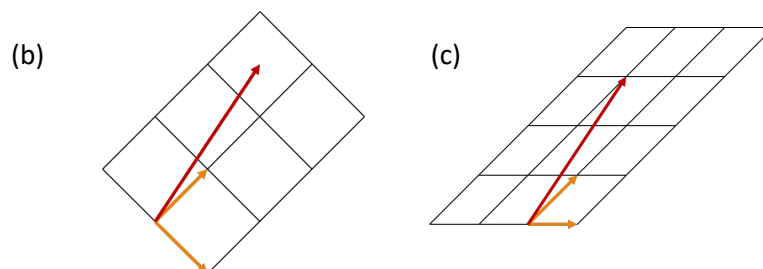
2.6.2 Standard VS Non-Standard Basis

Given an arrow that represents a certain vector in $\mathbb{R}^2$, how do we tell the coordinates of the vector? We need to set up the coordinate system as a reference for the units on both the x- and y-axis.



In this case, the coordinates of the vector (in red) is given by $(2,3)$, with respect to the standard coordinate system.

In some situation, we may need to re-orientate the coordinate system such as the following:



Then the same red arrow will have different coordinates with respect to the new coordinate systems.

How do we represent the different coordinate systems algebraically? We can do this using the various bases for $\mathbb{R}^2$.

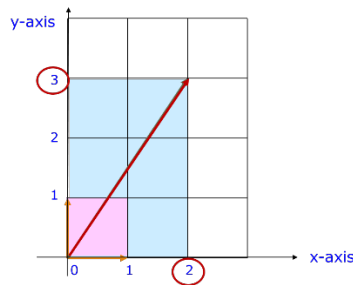
For the first coordinate system (a), it is given by the standard basis vectors, namely $(1, 0)$ and $(0, 1)$, represented by the two orange unit vectors along the x- and y- axis. In other words, the two unit vectors define the grid for the coordinate system.

The other two coordinate systems correspond to some non-standard bases for $\mathbf{R}^2$. The coordinate system (b) is given by the basis $(1, -1), (1, 1)$ as indicated by the two orange arrows, which gives us the new grid for the coordinate system. The coordinate system (c) is given by another basis $(1, 0), (1, 1)$ and hence a different shape for the grid.

Standard Basis S_1

$$S_1 = \{(1, 0), (0, 1)\}$$

The standard basis will give the unit block in pink, which in turn determines the unit of measurement along the x and y axis.



This will then give the x and y coordinates of our vector.

Algebraically, we can write $\mathbf{v} = (2, 3) = 2(1, 0) + 3(0, 1)$.

The scalars (coefficients) of this linear combination are 2 and 3. Putting these two numbers together, we get the coordinates of our vector $\mathbf{v}$ with respect to the standard basis S_1 :

$$(\mathbf{v})_{S_1} = (2, 3).$$

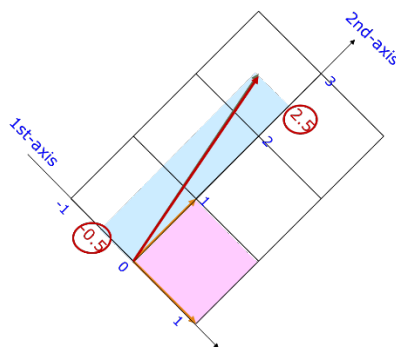
As expected, if we use the standard basis, the coordinates will be the same as the original one.

This is not the case if we use the non-standard basis.

Non-Standard Basis S_2

$$S_2 = \{(1, -1), (1, 1)\}$$

The grid, which is determined by the new basis vectors is given below, and we extend the coordinate axis, which we refer to as the 1st axis and 2nd axis, to differentiate them from the x- and y- axis. The basis gives the unit block in pink, which in turn determines the unit of measurement along the 1st and 2nd axis.



This time we see that the two coordinates of our vector become -0.5 and 2.5 respectively.

Algebraically, we can write $\mathbf{v} = (2, 3) = -\frac{1}{2}(1, -1) + \frac{5}{2}(1, 1)$.

The scalars of this linear combination are $-1/2$ and $5/2$ respectively. Putting these two numbers together, we get the new coordinates of $\mathbf{v}$ with respect to the non-standard basis S_2 :

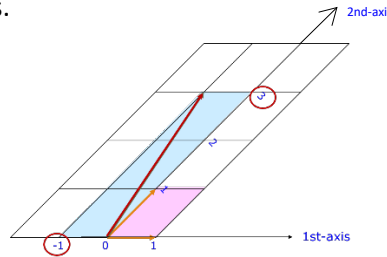
$$(\mathbf{v})_{S_2} = \left(-\frac{1}{2}, \frac{5}{2}\right).$$

We call this the coordinate vector of $\mathbf{v}$ relative to basis S_2 .

Non-Standard Basis S_3

$$S_3 = \{(1, 0), (1, 1)\}$$

This time the grid has a distorted shape, given by this basis, and the coordinate axis are not perpendicular to each other. The basis gives the unit block in pink, which in turns determine the unit of measurement along the two axis.



The two coordinates of our vector are -1 and 3 respectively.

Algebraically, we can write $\mathbf{v} = (2, 3) = -(1, 0) + 3(1, 1)$.

The scalars of this linear combination are -1 and 3 respectively. Once again, putting these two numbers together, we get the new coordinates of our vector $\mathbf{v}$ with respect to the non-standard basis S_3 :

$$(\mathbf{v})_{S_3} = (-1, 3)$$

This gives us the coordinate vector of $\mathbf{v}$ relative to basis S_3 .

2.6.3 Coordinate Vectors

We will now extend the notion of coordinate vectors to a general vector space.

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis for a vector space V and $\mathbf{v}$ a fixed vector in V .

We know that $\mathbf{v}$ can be written as a linear combination of this basis and in a unique way, as discussed earlier.

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k$$

Consider this unique set of scalars in the linear combination. We call them the coordinates of the vector $\mathbf{v}$ relative to the basis S . If we use these coordinates to form a k -vector, we will call this the coordinate vector of $\mathbf{v}$ relative to the basis S :

$$(\mathbf{v})_S = (c_1, c_2, \dots, c_k)$$

This coordinate vector $(\mathbf{v})_S$ is uniquely determined by the vector $\mathbf{v}$ and the basis S . It will change if we use a different basis for the same vector $\mathbf{v}$.

We should also take note that, the order of the vectors in the basis have to be fixed. If we change the order of some $\mathbf{u}_i$ and $\mathbf{u}_j$ in the basis, then the order of the coordinates will also change accordingly.

Example:

Let $S = \{ (1, 2, 1), (2, 9, 0), (3, 3, 4) \}$. Check that this is a basis for $\mathbf{R}^3$.

(a) Find the coordinate vector of $\mathbf{v} = (5, -1, 9)$ relative to S .

(b) Find a vector $\mathbf{w}$ in $\mathbf{R}^3$ such that $(\mathbf{w})_S = (-1, 3, 2)$.

The two questions are reverse of each other:

For (a), we are asking: given a vector $\mathbf{v}$, how do we find $(\mathbf{v})_S$?

For (b), we are asking: given coordinate vector $(\mathbf{w})_S$, how do we recover $\mathbf{w}$?

(a) We solve the equation $a(1, 2, 1) + b(2, 9, 0) + c(3, 3, 4) = (5, -1, 9)$.

By using Gaussian elimination, we get $a = 1, b = -1, c = 2$. i.e. $\mathbf{v} = 1(1, 2, 1) - (2, 9, 0) + 2(3, 3, 4)$.

So $(\mathbf{v})_S = (1, -1, 2)$.

(b) We have $(\mathbf{w})_S = (-1, 3, 2)$. So we substitute $a = -1, b = 3$ and $c = 2$ into

$$\mathbf{w} = a(1, 2, 1) + b(2, 9, 0) + c(3, 3, 4)$$

to get $\mathbf{w} = -1(1, 2, 1) + 3(2, 9, 0) + 2(3, 3, 4) = (11, 31, 7)$.

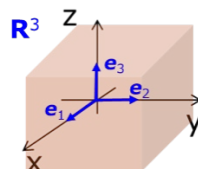
2.6.4 Meaning of Basis

Here's a summary of what we have discussed about bases for a vector space V .

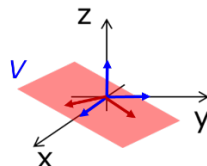
1. A basis for V is a building block of V
2. A basis for V is a "unit of measurement" for vectors in V .
3. A basis for V gives a "coordinate system" for V .

Note that a basis for " $\mathbf{R}^n$ " is not necessarily a basis for a "subspace of $\mathbf{R}^n$ "

For example, in the 3-space, we know that the standard basis vectors give a basis for $\mathbf{R}^3$.



But for a subspace V of $\mathbf{R}^3$ that is represented by a plane, the standard basis for $\mathbf{R}^3$ is not a basis for V . As shown in the diagram below, the vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ may not be on the plane itself.



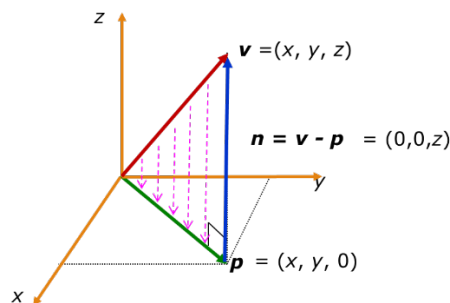
Instead, a basis for V should come from the subspace itself, and in this case, two vectors will be enough to generate V .

2.7 Projection and Linear Approximation

In this section, we discuss the concept of projection and linear approximation, which are related to dot product and the notion of orthogonality. In particular, we shall see how to find approximate solutions for systems that are inconsistent.

2.7.1 Projection of a Vector onto a Plane in $\mathbb{R}^3$

For a 3-vector $\mathbf{v} = (x, y, z)$ in $\mathbb{R}^3$, its projection onto the xy -plane is given geometrically as follow.



We drop the vector $\mathbf{v}$ (in red) perpendicularly onto the xy -plane to get the vector $\mathbf{p}$ (in green). The coordinate of $\mathbf{p}$ is $(x, y, 0)$. (The x and y coordinate is the same as $\mathbf{v}$, but the z coordinate is replaced by 0.)

If we complete the triangle as shown, this third side of the triangle is given by the vector $\mathbf{n}$ (in blue), which is equal to $\mathbf{v} - \mathbf{p}$. So its coordinate is $(0, 0, z)$. Note that this vector $\mathbf{n}$ is perpendicular to the xy -plane. Recall we also refer to perpendicular as orthogonal.

From this observation, we have:

$\mathbf{p}$ is a projection of $\mathbf{v}$ onto the plane	$\iff$	$\mathbf{v} - \mathbf{p}$ is orthogonal to the plane
--	--------	---

In fact, this holds not just for the xy -plane, but any plane in $\mathbb{R}^3$ that contains the origin.

We will further generalize the notion of projection to $\mathbb{R}^n$, with the plane replaced by a more general subspace of $\mathbb{R}^n$.

2.7.2 Projection of a Vector onto a Subspace of $\mathbb{R}^n$

Let V be a subspace of $\mathbb{R}^n$ and $\mathbf{u}$ a vector in $\mathbb{R}^n$.

If $\mathbf{p}$ is a vector in V such that $\mathbf{u} - \mathbf{p}$ is orthogonal to V , then $\mathbf{p}$ is called the projection of $\mathbf{u}$ onto V .

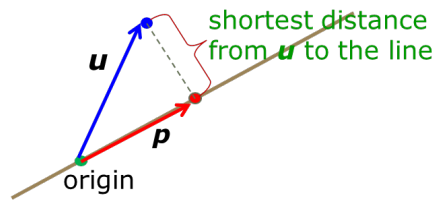
What do we mean by the **vector $\mathbf{u} - \mathbf{p}$ is orthogonal to a subspace V** ? This is defined to be:

$\mathbf{u} - \mathbf{p}$ is orthogonal to every vector $\mathbf{v}$ in V , i.e. the dot product $(\mathbf{u} - \mathbf{p}) \cdot \mathbf{v} = 0$.

Note that we always refer to a projection with respect to **certain subspace** of $\mathbb{R}^n$. Furthermore, every vector has **exactly one** projection onto a given subspace.

2.7.3 Projection and Best Approximation

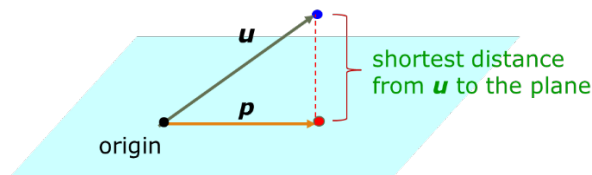
Suppose we are given a line and a point not on the line. Intuitively, the point on the line that is closest to the given point is the perpendicular drop onto the line. This can be expressed in terms of vectors:



$\mathbf{p}$ is the projection of $\mathbf{u}$ onto the line, and the distance between the vector $\mathbf{u}$ and its projection will give the shortest distance from $\mathbf{u}$ to the line.

So we say: $\mathbf{p}$ is the best approximation of $\mathbf{u}$ in the line.

Similarly, if we are given a plane and a point not on the plane. Like before, the point on the plane that is closest to the given point is the perpendicular drop onto the plane.



$\mathbf{p}$ is the projection of $\mathbf{u}$ onto the plane, and the distance between the vector $\mathbf{u}$ and its projection will give the shortest distance from $\mathbf{u}$ to the plane.

Again we say: $\mathbf{p}$ is the best approximation of $\mathbf{u}$ in the plane.

2.7.4 Best approximation of a Vector in a Subspace

The above observation can be generalized to any n-space.

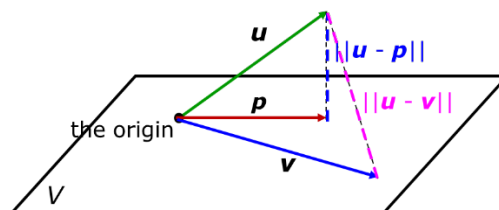
Let V be a subspace in $\mathbf{R}^n$ and $\mathbf{u} \in \mathbf{R}^n$. The subspace is not necessarily a line or a plane.

Before we continue, recall that for an n-vector $\mathbf{u}$, the norm of $\mathbf{u}$ is given by

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$$

Now let $\mathbf{p}$ be the projection of $\mathbf{u}$ onto V . Then

$$\|\mathbf{u} - \mathbf{p}\| \leq \|\mathbf{u} - \mathbf{v}\| \text{ for any vector } \mathbf{v} \text{ in } V.$$



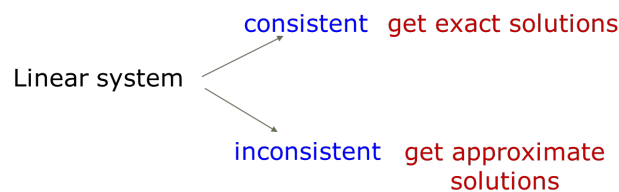
What the inequality says is that, among all the vectors $\mathbf{v}$ in the subspace V , the projection $\mathbf{p}$ is the one that is nearest to $\mathbf{u}$. i.e. $\mathbf{p}$ is the best approximation of $\mathbf{u}$ in V .

2.7.5 Approximate Solutions of Linear Systems

We have seen how the notion of projection and orthogonality can be applied to the study of best approximation. We shall now relate this to linear system.

We all know there are two types of systems, those that are consistent with solution, and those that are inconsistent without solution.

For consistent system, we have a systematic way to find the exact solutions. However, in real life, many linear systems that arise from mathematical modelling may not have exact solutions. For such inconsistent systems, we will find approximate solutions.



Example: Inconsistent System

Suppose there are three physical quantities r , s , t that are related by a quadratic equation:

$$t = cr^2 + ds + e \quad (*)$$

for some constant values c , d , e . We conduct an experiment to determine the exact values of c , d , e . To do this, we get the measurement for r , s , t from our experiments, substitute these readings into the quadratic equation $(*)$, and set up a linear system with c , d , e as variables.

Now suppose we repeat the experiment 6 times and obtained 6 sets of values as show in the table.

experiment # $\rightarrow$	i	1	2	3	4	5	6
input measurements $\rightarrow$	r_i	0	0	1	1	2	2
	s_i	0	1	2	0	1	2
output measurements $\rightarrow$	t_i	0.5	1.6	2.8	0.8	3.1	5.9

The top row indicates the experiment number, the second and third rows are our input measurements for r and s , while the bottom row gives the corresponding output measurement for t .

Like most experiments, while carrying out the measurements, there bound to be some experimental errors, that may lead to an inconsistent system.

Since there are 6 set of readings, we will have 6 equations, and this will form a 6×3 linear system as shown in the matrix equation form here.

$$\begin{array}{l}
 \text{6 equations} \\
 \text{3 unknowns } c, d, e \\
 \text{system } \mathbf{Ax} = \mathbf{b}
 \end{array}
 \rightarrow
 \begin{pmatrix}
 0 & 0 & 1 \\
 0 & 1 & 1 \\
 1 & 2 & 1 \\
 1 & 0 & 1 \\
 4 & 1 & 1 \\
 4 & 2 & 1
 \end{pmatrix}
 \begin{pmatrix}
 c \\
 d \\
 e
 \end{pmatrix}
 =
 \begin{pmatrix}
 0.5 \\
 1.6 \\
 2.8 \\
 0.8 \\
 5.1 \\
 5.9
 \end{pmatrix}$$

Note that the three columns of the coefficient matrix $\mathbf{A}$ correspond to the coefficients of c , d , and e respectively, and the constant matrix consists of the measurements of t .

Perform Gaussian elimination to this system to convince yourself that it is inconsistent. In other words, for any value we substitute for the column vector $\mathbf{x}$, $\mathbf{Ax}$ is not equal to $\mathbf{b}$, and hence

$$\mathbf{Ax} - \mathbf{b} \neq \mathbf{0}.$$

We shall see in a while how to find the “best approximate” solution for $\mathbf{Ax} = \mathbf{b}$. But for now, let us try to understand what do we mean by “best approximate” solution.

Since $\mathbf{Ax} - \mathbf{b}$ is not equal to zero, we want to find some value for $\mathbf{x}$ so that $\mathbf{Ax} - \mathbf{b}$ can be as close to $\mathbf{0}$ as possible.

In other words, we will try to find some vector $\mathbf{u}$ such that the norm $\|\mathbf{Au} - \mathbf{b}\|$ is the smallest.

Such a $\mathbf{u}$ is called a least squares solution to the system $\mathbf{Ax} = \mathbf{b}$.

2.7.6 Least Squares Solutions

A least squares solution of linear system $\mathbf{Ax} = \mathbf{b}$ is a vector $\mathbf{u}$ in $\mathbf{R}^n$ that minimize $\|\mathbf{Ax} - \mathbf{b}\|$.

$$\text{i.e. } \|\mathbf{Au} - \mathbf{b}\| \leq \|\mathbf{Av} - \mathbf{b}\| \text{ for all } \mathbf{v} \text{ in } \mathbf{R}^n$$

In other words, $\mathbf{Au}$ gives the best approximation for the vector $\mathbf{b}$ ($\mathbf{Au} \approx \mathbf{b}$).

From earlier discussion, this best approximation $\mathbf{Au}$ of $\mathbf{b}$ is given by the projection of $\mathbf{b}$ onto a certain subspace. We will discuss this in more details at the last part of this section. For now, we shall introduce a method to find least squares solutions of linear system $\mathbf{Ax} = \mathbf{b}$.

2.7.7 Finding Least Squares Solutions

- Starting with the system $\mathbf{Ax} = \mathbf{b}$ (1)
- Form the new linear system $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$ (2)
- Solve the system (2)
- A solution of (2) gives a least squares solution of (1)

System (2) $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$ is always consistent. It can have exactly one solution, or infinitely many solutions.

Example: Finding Least Squares Solutions

In our earlier experiment example:

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \\ 4 & 1 & 1 \\ 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} c \\ d \\ e \end{pmatrix} = \begin{pmatrix} 0.5 \\ 1.6 \\ 2.8 \\ 0.8 \\ 5.1 \\ 5.9 \end{pmatrix}$$

Denote this system by $\mathbf{Ax} = \mathbf{b}$ and recall that the system is inconsistent.

Now we consider the transpose of the coefficient matrix

$$\mathbf{A}^T = \begin{pmatrix} 0 & 0 & 1 & 1 & 4 & 4 \\ 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

and pre-multiply it to both $\mathbf{A}$ and $\mathbf{b}$ to get

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 34 & 14 & 10 \\ 14 & 10 & 6 \\ 10 & 6 & 6 \end{pmatrix} \text{ and } \mathbf{A}^T \mathbf{b} = \begin{pmatrix} 47.6 \\ 24.1 \\ 16.7 \end{pmatrix}$$

The new system $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$ is consistent and its solution $\mathbf{u}$ can be found (by Gaussian elimination):

$$\mathbf{u} = \begin{pmatrix} 0.9275 \\ 0.9225 \\ 0.3150 \end{pmatrix}$$

This vector $\mathbf{u}$ is the least squares solution of $\mathbf{Ax} = \mathbf{b}$, giving the best approximate solution.

2.7.8 Least Squares Solutions and Projection

As mentioned earlier, a least squares solution $\mathbf{u}$ of the linear system $\mathbf{Ax} = \mathbf{b}$ give the best approximation $\mathbf{Au}$ of $\mathbf{b}$. On the other hand, best approximation is closely related to projection. Indeed, it turns out that,

$$\mathbf{Au} = \mathbf{p} \text{ where } \mathbf{p} \text{ is the projection of } \mathbf{b} \text{ onto the column space* of } \mathbf{A}.$$

* The projection is on the column space of $\mathbf{A}$ because the product $\mathbf{Au}$ can be expanded as a linear combination of the columns of $\mathbf{A}$ (see below), which is in the column space of $\mathbf{A}$.

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \longleftrightarrow \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} x_1 + \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} x_2 + \cdots + \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} x_n$$

Since $\mathbf{p} = \mathbf{Au}$, it must belong to the column space of $\mathbf{A}$.

If we know a least squares solution $\mathbf{u}$ of $\mathbf{Ax} = \mathbf{b}$, we can find the projection $\mathbf{p}$ of $\mathbf{b}$ onto the column space of $\mathbf{A}$ simply by matrix multiplication $\mathbf{Au}$.

Example:

Find the projection of $(1,1,1,1)$ onto the subspace $V = \text{span}\{(1,-1,1,-1), (1,2,0,1), (2,1,1,0)\}$ of $\mathbb{R}^4$.

First, we need to set up a linear system and find its least squares solutions.

We use the three vectors in the linear span of V to form a 4×3 matrix $\mathbf{A}$ and the vector $(1, 1, 1, 1)$ to be $\mathbf{b}$ in column form:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Note that V is the column space of $\mathbf{A}$.

Now we have a linear system $\mathbf{Ax} = \mathbf{b}$. We find its least squares solution by solving $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$.

The general solution of the new system is given by $\mathbf{x} = \begin{pmatrix} -t + \frac{2}{5} \\ -t + \frac{4}{5} \\ t \end{pmatrix}$

Note that there are infinitely many solutions for this system. All of them are least squares solutions of $\mathbf{Ax} = \mathbf{b}$. We just need one particular solution, say let $t = 0$, which gives

$$\mathbf{u} = \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix}$$

Then $\mathbf{Au} = \begin{pmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{6}{5} \\ \frac{6}{5} \\ \frac{2}{5} \\ \frac{2}{5} \end{pmatrix}$ will give the required projection onto the subspace V .